

GRPO-MMS: Reinforcement Learning with a Composite Reward for Multimodal Fraud Message Detection

Zhengchao Huang^a, Jiaran Li^a, Wenming Yang^{a,*}

^aShenzhen International Graduate School, Tsinghua University, Shenzhen, 518055, China

Abstract

As text-only interception matures, fraudsters increasingly hide deceptive content in images, producing multimodal scams that evade text-based filters. Existing fraud detectors target pure-text short messages (SMS) and lack mechanisms for joint image-text reasoning. We propose **GRPO-MMS**, a reinforcement-learning framework based on Group Relative Policy Optimization (GRPO) and driven by a composite reward for multimodal fraud detection. Three contributions are made. (i) We release **MMS-VQA**, a multimodal fraud dataset of 2,086 image-text pairs spanning eight categories and three input modalities (text-only, image-only, image-text), with a generator-based split for cross-domain evaluation. (ii) We design a three-component composite reward combining format, authenticity and scene rewards, where the scene reward injects fine-grained fraud-type gradients into GRPO advantage estimation. (iii) On Qwen2.5-VL-3B and 7B, GRPO yields up to +31.36 percentage points in 8-class accuracy and +21.39 pp in Macro-F1 on the image-text modality. Ablations show that a minimal scene-reward weight ($\lambda=0.05$) is sufficient to activate fine-grained classification, achieving 81.50% 8-class accuracy. Results highlight the complementary value of vision and language and the effectiveness of reward shaping for multimodal pattern recognition.

*Corresponding author.

Email addresses: huang-zc23@mails.tsinghua.edu.cn (Zhengchao Huang),
lijiaran24@mails.tsinghua.edu.cn (Jiaran Li), yang.wenming@sz.tsinghua.edu.cn
(Wenming Yang)

Keywords: Multimodal fraud detection, Reinforcement learning, Reward shaping, Visual question answering, Multimodal large language model

1. Introduction

Telecommunication fraud has emerged as one of the most pervasive threats to personal information security worldwide. Traditional text-based detectors [1, 2] have filtered a large volume of scam short messages (SMS), yet as text-only interception matures, fraudsters are rapidly migrating toward multimodal attack vectors. By embedding deceptive content in images—fabricated official notices, phishing posters, or synthetic scene photographs—and pairing them with persuasive text, attackers simultaneously circumvent text-only filters and enhance the perceived authenticity of their messages [3, 4].

This evolution is accelerated by the rapid progress of text-to-image generation, including Stable Diffusion [5], DALL·E 2 [6], FLUX [7] and Seedream [8], which produce highly realistic synthetic imagery. When combined with carefully crafted text, multimodal scams exhibit substantially greater deceptive power than their text-only counterparts: even cautious users struggle to distinguish authentic communications from fabricated ones.

Despite the growing threat, research on fraud detection remains largely unimodal—focusing on either text classification or image forensics—with limited attention to the joint image–text scenario. Multimodal fake-news detection [9, 10, 11, 12] and deep-fake detection [13, 14] are related yet distinct, because fraud messages exhibit fraud-specific linguistic patterns and visual cues that differ from general misinformation. Moreover, most existing methods rely on supervised paradigms that require extensive labels and struggle to inject structured reasoning about fraud types.

Recent advances in multimodal large language models (MLLMs) such as GPT-4V [15], LLaVA [16] and Qwen2.5-VL [17] open new avenues for this task. These models process interleaved image–text inputs, perform visual question answering (VQA) [18],

and generate structured reasoning chains through chain-of-thought (CoT) prompting [19]. In parallel, reinforcement-learning techniques such as RLHF [20] and Group Relative Policy Optimization (GRPO) [21] provide efficient frameworks for aligning model behaviour with domain-specific objectives through carefully designed reward signals.

Building on these developments, we propose **GRPO-MMS**, a reinforcement-learning framework for multimodal fraud message detection. Our approach addresses three key challenges:

1. **Dataset construction.** We build **MMS-VQA**, a multimodal fraud dataset. Starting from a public Chinese telecom fraud SMS corpus [11], we apply keyword-based filtering to extract high-risk scam samples, retrieve real images for legitimate messages via TF-IDF keywords and web image search, and synthesise fraud images using five state-of-the-art generators. The dataset covers eight categories and three input modalities (text-only, image-only, image-text); a generator-based train/test split evaluates cross-domain generalisation.
2. **Composite reward design.** We introduce a three-component reward $R = R_{\text{format}} + (1 - \lambda)R_{\text{accuracy}} + \lambda R_{\text{scene}}$ that jointly supervises output format, binary authenticity judgement and fine-grained fraud-type classification. The scene reward injects an auxiliary semantic signal that enriches GRPO advantage estimation without overwhelming the primary detection objective.
3. **Comprehensive evaluation.** On Qwen2.5-VL-3B and 7B, GRPO-MMS produces substantial improvements across all metrics. The image-text modality benefits the most (Macro-F1: 44.88%→66.27% on 7B). A dedicated ablation on λ reveals a pronounced non-linear response, in which even $\lambda=0.05$ triggers a dramatic jump in scene-classification performance.

The remainder of this paper is organised as follows. Section 2 reviews related work; Section 3 describes the MMS-VQA dataset; Section 4 presents the GRPO-based

composite reward; Section 5 reports experimental results, ablations and qualitative analyses; Section 6 concludes.

2. Related Work

2.1. Text-based fraud and scam detection

Telecom fraud detection has been extensively studied for more than a decade, evolving from rule-based keyword matching and classical machine learning to deep neural networks. Early methods relied on handcrafted features—bag-of-words, TF-IDF and n -grams—combined with support vector machines or random forests for SMS spam and smishing classification [4, 3]. With the advent of deep learning, recurrent and convolutional networks were adopted to capture sequential and local patterns in text, noticeably improving detection rates.

More recently, pre-trained language models have been applied to fraud detection. BERT-based approaches [2] leverage contextual embeddings to capture the semantic nuances of fraudulent text. Jiang et al. [11] proposed a neuron-selection-based method that taps large language models for telecom fraud recognition, further demonstrating the potential of LLMs in this domain. These methods, however, are inherently limited to textual inputs. As fraudsters increasingly embed deceptive content in images—yielding multimodal phishing messages, counterfeit promotional materials and AI-generated scam posters—text-only pipelines no longer provide sufficient coverage, motivating multimodal detectors that jointly analyse visual and textual content.

2.2. Multimodal fake news and misinformation detection

Multimodal fake-news detection targets the identification of manipulated image–text pairs on social media and news platforms, and is closely related to fraud detection. Zhou et al. [9] proposed the SAFE framework, which measures similarity-aware correspondence between visual and textual features and treats semantic discrepancy as a manipulation indicator. Wu et al. [10] developed a co-attention network (MCAN) that

models fine-grained inter-modal interactions. Zhou et al. [11] proposed FND-CLIP, exploiting CLIP-aligned features to guide cross-modal fusion. Chen et al. [12] formulated cross-modal ambiguity learning from an information-theoretic perspective, handling cases where image–text consistency is deliberately obscured. Wei et al. [22] investigated cross-modal knowledge distillation (CMC) to transfer discriminative features between modalities when one modality provides weak signals, while Jing et al. [23] introduced progressive fusion networks that integrate multimodal features at multiple semantic levels.

While these methods advance multimodal misinformation detection, they focus on general news content rather than fraud scenarios. Fraud messages differ from fake news in three important respects: (i) they exhibit structured category taxonomies (e.g., impersonating law enforcement, fake loans, phishing) requiring fine-grained scene classification rather than binary true/false judgement; (ii) their visual component is typically AI-generated from scratch rather than misappropriated from other contexts; and (iii) the detector should provide interpretable reasoning over fraud types to enable targeted downstream responses. Our work addresses these fraud-specific challenges.

2.3. Deepfake and AI-generated content detection

The proliferation of AI-generated images has catalysed significant work on deepfake detection. Rana et al. [13] provide a systematic review categorising approaches by the forgery artefacts they exploit (frequency, spatial and temporal cues). Wang et al. [14] proposed an adversarial-game framework for deepfake forensics that co-trains the detector and the generator, improving robustness against unseen manipulations. Miao et al. [24] introduced hierarchical frequency-assisted interactive networks (HFI-Net), demonstrating the importance of multi-scale frequency analysis for detecting subtle generation artefacts. More recently, Shi et al. [25], published in *Pattern Recognition*, proposed a real-face foundation representation that learns a generalisable prior from pristine faces and achieves strong cross-dataset transfer to unseen deepfake

sources.

These methods focus primarily on visual forgery detection—determining whether an image is real or AI-generated—without considering the accompanying textual context. In multimodal fraud messages, the image is one component of a larger deceptive narrative, and a detector must not only assess image authenticity but also integrate visual evidence with textual analysis to determine the overall intent and category of the message. We therefore formulate the task as a VQA problem in which the model must jointly reason over both modalities.

2.4. Multimodal large language models

The emergence of MLLMs has dramatically expanded joint vision–language understanding. GPT-4V [15] showcased strong visual reasoning across diverse tasks. LLaVA [16] pioneered visual instruction tuning by connecting a vision encoder to an LLM through a lightweight projection. BLIP-2 [26] bridged frozen image encoders and LLMs through a querying Transformer. Mini-Gemini [27] explored dual vision encoders to mine the potential of multi-modality, and Qwen2.5-VL [17] offered strong multilingual support together with competitive multimodal benchmarks, making it well-suited for Chinese-language detection. In parallel, MLLMs have been increasingly coupled with knowledge prompting for domain adaptation: Hu et al. [28] published in *Pattern Recognition* prompted LLMs with context and a pre-answer for knowledge-based VQA, while Alawwad et al. [29], also in *Pattern Recognition*, combined LLMs with retrieval-augmented generation for textual textbook question answering.

Beyond MLLMs, multimodal learning has been surveyed as a broader research area. Baltrušaitis et al. [30] proposed an influential taxonomy covering representation, translation, alignment, fusion and co-learning. CLIP [31] demonstrated that contrastive pre-training on web-scale image–text pairs yields transferable visual representations aligned with language. Xu et al. [32] surveyed multimodal learning with

Transformers, covering fusion strategies, pre-training objectives and downstream applications. In addition, Zhang et al. [53], published in *Pattern Recognition*, proposed a self-adaptive image–text fusion scheme that dynamically weights the two modalities to address their semantic gap—a perspective that is also relevant when reasoning over image–text fraud messages.

A key challenge for MLLMs is that zero-shot or instruction-tuned performance on domain-specific tasks often falls short of deployment requirements [54]. In our setting the difficulty is compounded by the need to perform binary classification (authentic vs. fraudulent) *and* fine-grained scene categorisation simultaneously—a dual-objective problem that standard instruction tuning cannot efficiently address.

2.5. Reinforcement learning for language model alignment

Reinforcement learning from human feedback (RLHF) [20] and its variants have become the dominant paradigm for aligning language models with desired behaviour. The standard RLHF pipeline trains a reward model from human preferences and optimises the language model with Proximal Policy Optimization (PPO) [55]. Although effective, PPO requires a separate value network (critic), which raises computational and memory overhead.

GRPO [21], introduced by DeepSeek-Math, dispenses with the value network by sampling a group of G candidate outputs per query and deriving advantages through within-group reward normalisation. This paradigm has proven effective on mathematical reasoning and has shown promise for other structured-generation tasks. DeepSeek-R1 [56] further demonstrated that RL training with carefully designed rewards can incentivise sophisticated reasoning behaviour in LLMs, including chain-of-thought deliberation, without explicit supervision.

The design of reward functions is critical for RL-based optimisation. Multi-component rewards that combine format compliance with task-specific accuracy signals have been shown to benefit structured generation [21, 56]. Existing applications, however, pri-

marily target mathematical reasoning or general alignment. We extend this paradigm to the multimodal fraud detection domain by introducing a *scene reward* that supplies fine-grained semantic supervision for fraud-type classification. Unlike prior work that relies on purely binary rewards, our composite reward creates a multi-granularity optimisation landscape that simultaneously drives detection accuracy and category-level discriminability, enabling the model to learn fraud-specific reasoning patterns—for instance, treating requests to install screen-sharing applications as indicators of remote-control scams—that cannot be acquired by supervised fine-tuning alone.

3. The MMS-VQA Dataset

Fig. 1 illustrates the overall pipeline for constructing MMS-VQA. It follows two paths: real-image retrieval for legitimate messages and synthetic-image generation for fraud messages, both of which we now describe.

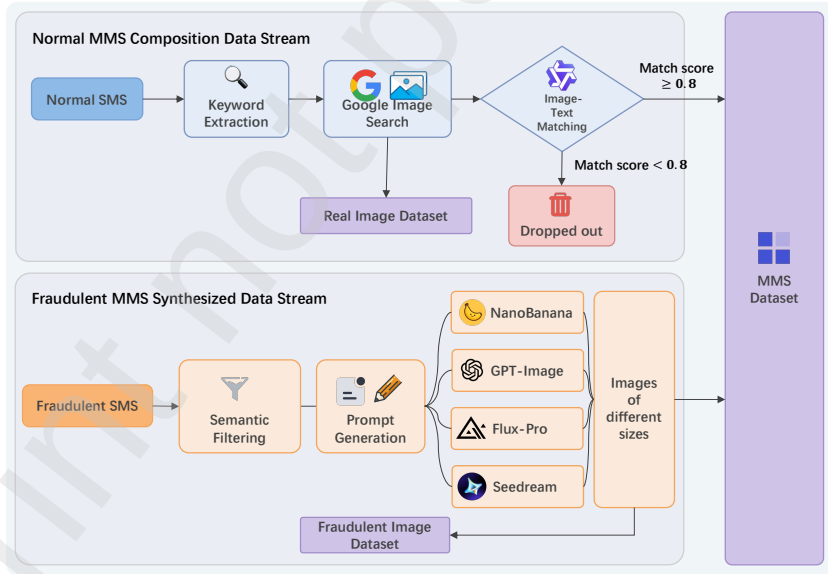


Fig. 1: Overview of the MMS-VQA construction pipeline. Legitimate SMS are paired with retrieved real images, whereas fraud SMS are combined with AI-generated images to form multimodal samples.

3.1. Fraud text collection and filtering

Our text data originate from a public Chinese telecom fraud SMS dataset [10] containing 28,357 messages across eight categories: legitimate text and seven fraud types (impersonating law enforcement, fake loans, impersonating customer service, impersonating leaders/acquaintances, fake task rewards, romance scams and gaming transaction fraud). The original distribution is shown in Fig. 2(a), where fake task rewards dominate (24.8%) and romance scams are least frequent (5.5%).

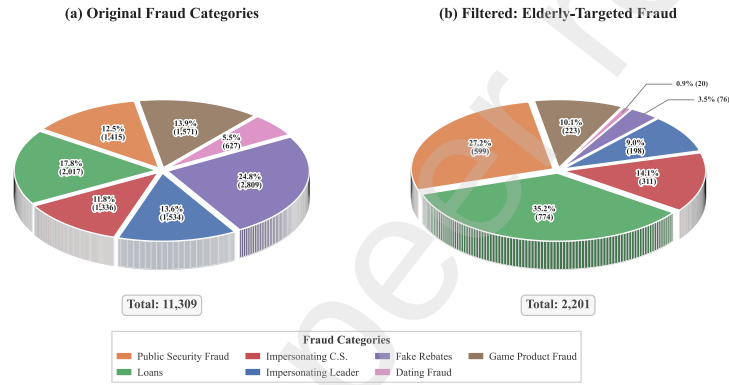


Fig. 2: Distribution of fraud categories before (a) and after (b) keyword-based filtering. Fake loans and impersonating law enforcement dominate after filtering, reflecting the most prevalent high-risk fraud scenarios.

Because not all fraud types are equally representative of high-risk scenarios, we design keyword-based filtering rules covering nine prevalent scam topics: (1) medical/social-insurance fraud; (2) health- or epidemic-related fraud; (3) pension/financial fraud; (4) law-enforcement impersonation demanding a “safe-account” transfer; (5) exploitation of concern for family members; (6) savings-account fraud; (7) authority-pressure tactics (e.g., fabricated arrest warrants); (8) remote-control/screen-sharing scams; and (9) bank- or financial-institution impersonation.

After filtering, all 17,048 legitimate messages are retained and 2,201 high-risk fraud messages are extracted. As shown in Fig. 2(b), fake loans (35.2%) and law-

enforcement impersonation (27.2%) together account for 62.4% of the filtered data. Notably, fake task rewards drop from 24.8% to 3.5%, whereas law-enforcement impersonation rises from 12.5% to 27.2%, reflecting the prevalence of authority-based scam tactics.

3.2. Image retrieval and generation

Real image retrieval for legitimate messages. For legitimate SMS we adopt a keyword-based retrieval strategy (Fig. B): core keywords are extracted per message using TF-IDF and submitted as queries to a commercial image search service, yielding three candidate images per message. All retrieved images enter the real-image pool. Qwen2.5-VL-32B then scores image-text matches; only pairs with scores ≥ 0.8 are kept, ensuring semantic consistency.

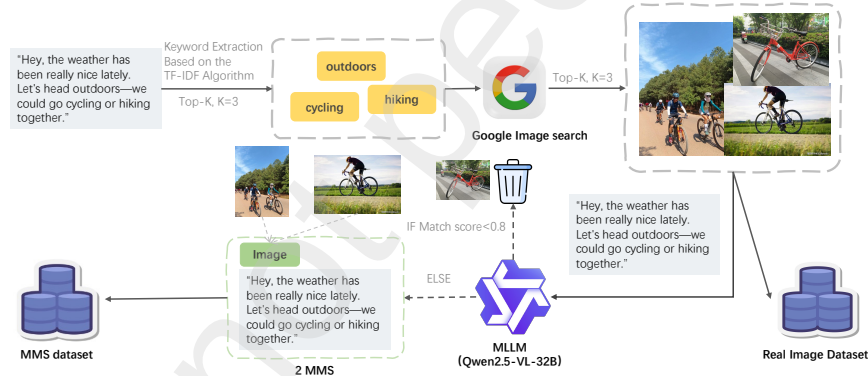


Fig. 3: Pipeline for synthesising multimodal samples from legitimate SMS via keyword-based image retrieval.

Synthetic image generation for fraud messages. For fraud SMS we leverage multiple state-of-the-art text-to-image models to generate matching images (Fig. D). Each fraud message is processed by the following steps: (1) a template type is sampled uniformly (poster or scene); (2) one of five prompt templates is randomly selected for the chosen type and the fraud text is embedded within it; (3) one of five generators—{NanoBanana2, NanoBananaPro, FLUX 2 Pro [7], GPT-Image 1.5, See-

dream 4.5 [8])—is randomly invoked to synthesise the image. The use of five stylistically diverse generators is crucial for training robust detectors [5, 6].

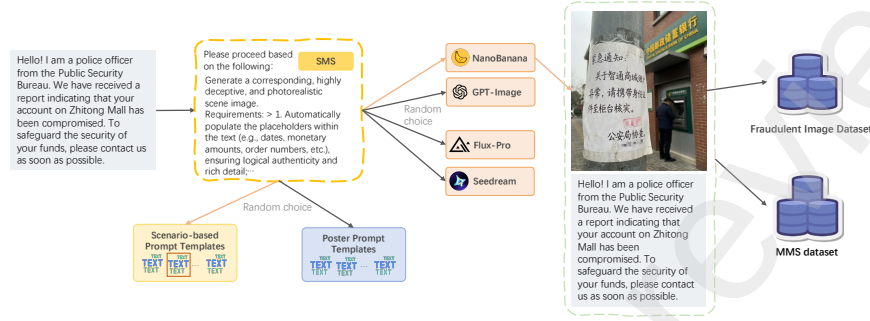


Fig. 4: Pipeline for synthesising multimodal samples from fraud SMS via AI image generation.

3.3. Dataset statistics and VQA formulation

The resulting MMS dataset (Fig. 5) comprises 1,681 real samples (4,664 images) and 807 generated samples (807 images). Among the generated samples, NanoBanana2 contributes 215, GPT-Image 1.5 contributes 200, Seedream 4.5 contributes 169, NanoBananaPro contributes 118, and FLUX 2 Pro contributes 105; scene-type images (441) outnumber poster-type images (366).

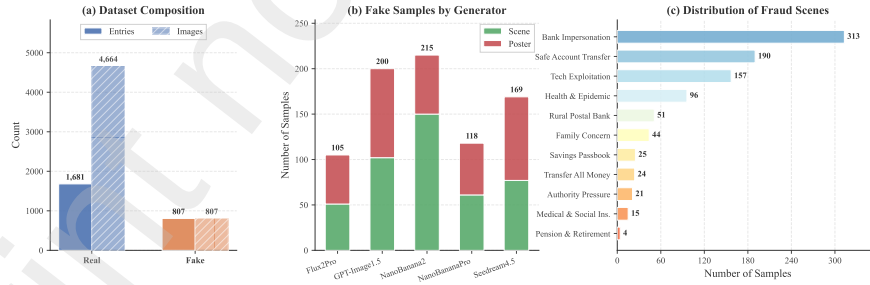


Fig. 5: Statistics of the MMS fraud-detection dataset, including the distribution of real and generated images across generators and template types.

To enable VQA-style training [18], each sample is converted into one of three question types: (i) **image-text**, in which both image and SMS text are provided; (ii)

Table 1: Representative examples from the three input modalities in the MMS-VQA dataset. Text-only samples contain no image (marked “—”); image-only samples provide only the visual input; image-text samples combine both. Some text content is abbreviated for space.

Modality	Image	Question	Answer
Text-only	—	Is this a scam or a legitimate message? “Hello, this is the JD Finance customer service. We found that your JD credit account has a compliance violation...”	Impersonating customer service
	—	Is the following a scam SMS? “Huarun PLACE community store year-end sale... Spend over XX yuan and receive a lucky bag.”	Legitimate
Image-only		Analyse the image and determine whether it is a scam. Identify the fraud type.	Impersonating law enforcement
		Based on the image, is this legitimate or telecom fraud?	Legitimate
Image + Text		Is this legitimate or fraudulent? “Dear Grandpa, I am your grandson. I got into trouble abroad... I need you to help me pay the compensation...”	Impersonating leaders
		Is this a scam SMS? “Hey, the weather has been great recently. Let’s go for outdoor activity—cycling or hiking.”	Legitimate

image-only, in which only the image is provided; and (iii) **text-only**, in which only the SMS text is provided. Table 1 presents representative examples for each modality, illustrating the diversity of input formats and their ground-truth labels. In the text-only setting the model must rely entirely on linguistic cues to detect fraud; in the image-only setting it must extract visual evidence (e.g., synthetic poster text, fabricated logos) in isolation; in the image-text setting it must integrate both modalities to form a joint judgement.

The training set adopts a 5:4:1 ratio (image-text : image-only : text-only) with balanced classes (1:1 legitimate : fraud). A *generator-based split* is employed: all 118 NanoBananaPro-generated fraud samples are held out for testing to evaluate cross-generator generalisation. An equal number of legitimate samples is added to preserve

class balance, and all three modalities are generated for every test sample.

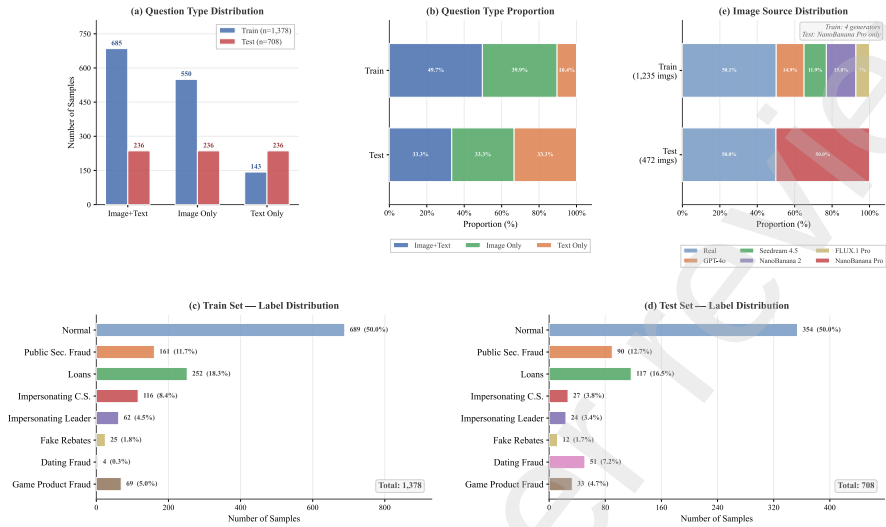


Fig. 6: Statistics of the MMS-VQA dataset: (a–b) modality distribution in the training and test sets; (c–d) category distribution; (e) image-source distribution.

The final MMS-VQA dataset (Fig. 6) comprises 1,378 training and 708 test samples. The training set is dominated by image–text (49.7%) and image-only (39.9%) samples, whereas the test set is modality-balanced (~33% each). Importantly, the training images come from four generators plus real sources, whereas the test images come only from real sources and the held-out NanoBananaPro generator, providing a rigorous evaluation of cross-domain generalisation. The MMS-VQA dataset thus combines modality-level balance on the test side with generator-level openness, which together stress-test both cross-modal reasoning and cross-generator robustness.

4. Method: GRPO with a Composite Reward

4.1. GRPO objective and advantage estimation

We adopt GRPO [21] as our RL training algorithm. Unlike standard PPO [35], which requires a separate value network for baseline estimation, GRPO samples a

group of G candidate outputs per input query and estimates relative advantages through within-group reward normalisation. The GRPO objective is:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(O|q)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(\frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q, o_{i,<t})} \hat{A}_{i,t}, \text{clip}(\cdot, 1-\epsilon, 1+\epsilon) \hat{A}_{i,t} \right) - \beta D_{\text{KL}}[\pi_{\theta} \parallel \pi_{\text{ref}}] \right], \quad (1)$$

where q is the input query, G is the group size, o_i is the i -th candidate output, ϵ is the clipping range, β is the KL penalty coefficient, and π_{ref} is the reference policy. The advantage estimate $\hat{A}_{i,t}$ is computed as:

$$\hat{A}_{i,t} = \frac{R(o_i) - \text{mean}(\{R(o_j)\}_{j=1}^G)}{\text{std}(\{R(o_j)\}_{j=1}^G)}, \quad (2)$$

where $R(o_i)$ is the reward for the i -th candidate output. Outputs with above-average rewards receive positive advantages and are reinforced during training. The quality of the reward function R therefore critically determines the optimisation direction and convergence behaviour.

4.2. Composite reward function

For fraud-message detection the model must perform two tasks simultaneously: (i) binary authenticity judgement (legitimate vs. fraudulent) and (ii) fine-grained fraud-type classification. The latter is important for interpretability—distinguishing “impersonating law enforcement” from “fake loans” enables targeted downstream responses—and provides richer semantic supervision during training.

Inspired by the structured-output design of DeepSeek-R1 [36], we require the model to produce outputs in the following format:

```
<think>reasoning process</think>
<authenticity>legitimate/fraudulent</authenticity>
<scene>fraud type</scene>
```

The <think> tag elicits chain-of-thought reasoning [19], while <authenticity> and <scene> capture the binary and fine-grained predictions, respectively.

The total reward R combines three components:

$$R = R_{\text{format}} + (1 - \lambda) R_{\text{accuracy}} + \lambda R_{\text{scene}}, \quad (3)$$

where $\lambda \in [0, 1)$ balances accuracy and scene rewards.

Format reward. R_{format} checks whether the output contains all three required tags (<think>, <authenticity>, <scene>) in the correct order via regular-expression matching:

$$R_{\text{format}} = \begin{cases} 0, & \text{if the output matches the required format;} \\ -1, & \text{otherwise.} \end{cases} \quad (4)$$

Setting the compliant reward to 0 (rather than a positive value) prevents format compliance from dominating the reward signal, permitting clearer observation of other reward components.

Accuracy reward. R_{accuracy} evaluates the binary authenticity judgement:

$$R_{\text{accuracy}} = \begin{cases} 1, & \text{if authenticity matches the ground truth;} \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

This constitutes the primary supervision signal, directly driving detection accuracy.

Scene reward. R_{scene} evaluates fraud-type classification:

$$R_{\text{scene}} = \begin{cases} 1, & \text{if the predicted fraud type matches the ground truth;} \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

Without the scene reward ($\lambda = 0$), the model must only classify messages as legitimate or fraudulent. With $\lambda > 0$, the model must further identify the specific fraud category to maximise its reward, encouraging a deeper understanding of category-specific semantic features rather than shallow statistical shortcuts.

The constraint $\lambda < 1$ guarantees that the accuracy reward always retains a non-zero weight $(1 - \lambda) > 0$, preventing the auxiliary scene signal from overwhelming the primary detection task. The total reward ranges over $[-1, 1]$: $R = 1$ when all three components are maximised, and $R = -1$ when the format is violated and both classifications are incorrect.

5. Experiments

5.1. Experimental setup

Data split. The MMS-VQA training set is further divided into training and validation subsets with an 8:2 ratio. The test set (708 samples) remains untouched.

Models. We evaluate two model scales, Qwen2.5-VL-3B-Instruct and Qwen2.5-VL-7B-Instruct [17], both instruction-tuned MLLMs with joint image-text understanding capability.

Training. We use verl [37] as the RL training framework with GRPO optimisation. Key hyperparameters are as follows: batch size 128; learning rate 1×10^{-6} ; 5 epochs (~ 40 training steps); group size $G = 5$; PPO mini-batch size 32; KL coefficient $\beta = 0.01$; maximum prompt length 3840 tokens; maximum response length 1024 tokens. Both 3B and 7B models use identical configurations to ensure a fair comparison. The scene-reward weight is fixed at $\lambda = 0.1$ unless otherwise specified.

Metrics. We report: (i) 2-class accuracy (2-Cls Acc); (ii) 2-class F1; (iii) 8-class accuracy (8-Cls Acc); and (iv) 8-class Macro-F1. The 8-class metrics assess fine-grained recognition over seven fraud types plus legitimate text.

5.2. Training dynamics and overall performance

Fig. 7 shows the training dynamics over 40 steps. Both models exhibit stable convergence, with systematic differences attributable to model capacity. The 7B model starts from a higher initial reward (0.59 vs. 0.23 for 3B) and converges in approxi-

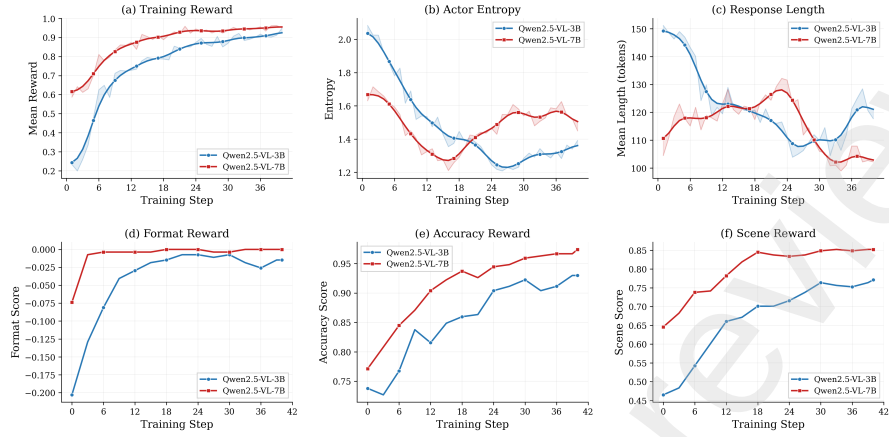


Fig. 7: Training dynamics of GRPO over 40 steps. (a) Training reward; (b) policy entropy; (c) response length; (d) format reward; (e) accuracy reward; (f) scene reward. The 7B model converges faster and attains higher final performance than the 3B model.

Table 2: Performance comparison with state-of-the-art VLMs and GRPO-trained Qwen2.5-VL baselines on the MMS-VQA test set. Improvements in parentheses are reported relative to the corresponding Qwen2.5-VL baseline. Best results are in **bold**.

Model	Binary (2-class)		Fine-grained (8-class)	
	Acc	F1	Acc	Macro-F1
Gemini-3.1-Pro-Preview	91.24	93.82	74.01	59.55
GPT-5.4	95.06	95.06	75.99	57.11
Qwen2.5-VL-3B	80.51	80.55	46.89	29.78
Qwen2.5-VL-3B + GRPO	96.47 ^{+15.96}	96.54 ^{+15.99}	78.25 ^{+31.36}	45.13 ^{+15.35}
Qwen2.5-VL-7B	79.66	79.95	64.12	45.95
Qwen2.5-VL-7B + GRPO	97.46 ^{+17.80}	97.53 ^{+17.58}	83.47 ^{+19.35}	61.18 ^{+15.23}

mately 20 steps, whereas the 3B model requires the full 40 steps. This reflects the 7B model’s richer semantic understanding acquired during instruction tuning.

Policy entropy (Fig. 7b) reveals different exploration–exploitation trade-offs: the 3B model’s entropy monotonically decreases, indicating progressive policy consolida-

tion, while the 7B model exhibits a V-shaped trajectory, recovering some exploration after reaching minimum entropy. This suggests that larger models preserve generation diversity while establishing stable decision policies, helping to avoid premature policy collapse.

Format reward (Fig. 4d) converges fastest; the 7B model eliminates format violations by step 6, indicating that structured-output compliance is a low-difficulty adaptation task for pretrained MLLMs. Accuracy and scene rewards converge more slowly, with the scene reward starting from a much lower initial value—consistent with fine-grained classification being inherently more challenging than binary detection.

Table 2 summarises the test-set results, where we additionally benchmark two state-of-the-art proprietary multimodal LLMs—Gemini-3.1-Pro-Preview and GPT-5.4—under a zero-shot prompting setting that follows the same structured output format used by our Qwen2.5-VL pipeline. Both GRPO-trained Qwen2.5-VL baselines achieve 16–18 pp improvements in 2-class accuracy, and the 8-class gains are even larger (+31.36 pp for 3B and +19.35 pp for 7B). The disproportionately larger 8-class gains relative to 2-class gains confirm that the scene reward provides effective fine-grained supervision. Interestingly, the 3B model shows a larger absolute 8-class gain (+31.36 vs. +19.35), reflecting its lower baseline (46.89% vs. 64.12%) and greater optimisation headroom.

The comparison with the proprietary models offers two additional observations. First, zero-shot GPT-5.4 and Gemini-3.1-Pro-Preview deliver strong baselines (2-class accuracy 95.06% and 91.24%, respectively), substantially surpassing the untrained Qwen2.5-VL-3B/7B models and confirming that the MMS-VQA task is non-trivial yet tractable for top-tier MLLMs. Second, after GRPO training our 7B model still outperforms both proprietary systems by a clear margin on every metric—+2.40 pp in 2-class accuracy and +7.48 pp in 8-class accuracy over GPT-5.4, and +1.63 pp in 8-class Macro-F1 over Gemini-3.1-Pro-Preview, the strongest proprietary competitor on fine-grained classification. This indicates that a compact 7B open-source MLLM, when

aligned with a properly designed composite reward, can surpass much larger closed-source models on domain-specific fraud detection, which is particularly attractive for privacy-sensitive deployment. We also note that the two proprietary models exhibit complementary profiles: GPT-5.4 is stronger on binary authenticity judgement, whereas Gemini-3.1-Pro-Preview attains a higher 8-class Macro-F1, suggesting different trade-offs between detection confidence and category-level discriminability—an asymmetry that GRPO-MMS resolves through its composite reward.

5.3. Category-level and modality-level analysis

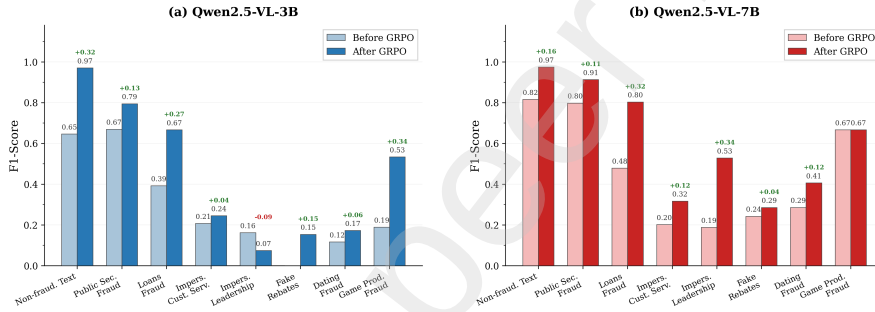


Fig. 8: Per-category F1 scores before and after GRPO training. High-frequency categories show consistent gains, while low-frequency categories benefit more from larger model capacity.

Fig. 8 shows per-category F1 scores. High-frequency categories (legitimate text, law-enforcement impersonation, fake loans) improve consistently on both models, with legitimate text reaching a near-saturation F1 of 0.97. Low-frequency categories exhibit capacity-dependent behaviour: the 3B model suffers an F1 degradation on impersonating leaders/acquaintances ($0.16 \rightarrow 0.07$) because the limited training samples in that category yield insufficient reward-signal diversity for effective GRPO optimisation. The 7B model, benefiting from its larger parameter space, attains positive gains even on rare categories (impersonating leaders/acquaintances: $0.19 \rightarrow 0.53$).

Table 3 shows modality-level results. A noteworthy finding is the asymmetric gain pattern: the image-only task has a higher binary baseline than text-only (3B: 82.63%

Table 3: Performance comparison across input modalities before and after GRPO training.

Modality	Model	2-Class		8-Class	
		Acc	F1	Acc	Macro-F1
Text-only	Qwen2.5-VL-3B	77.54	77.03	47.46	29.24
	Qwen2.5-VL-3B + GRPO	96.61 +19.07	96.81 +19.78	81.36 +33.90	54.99 +25.75
	Qwen2.5-VL-7B	78.81	77.98	63.14	51.41
	Qwen2.5-VL-7B + GRPO	97.46 +18.65	97.46 +19.48	83.90 +20.76	64.06 +12.65
Image-only	Qwen2.5-VL-3B	82.63	82.82	55.93	29.74
	Qwen2.5-VL-3B + GRPO	94.92 +12.29	94.91 +12.09	75.85 +19.92	33.16 +3.42
	Qwen2.5-VL-7B	86.02	86.30	70.34	39.12
	Qwen2.5-VL-7B + GRPO	96.19 +10.17	96.39 +10.09	80.51 +10.17	52.06 +12.94
Image+Text	Qwen2.5-VL-3B	81.36	80.80	37.29	28.87
	Qwen2.5-VL-3B + GRPO	97.88 +16.52	97.88 +17.08	77.54 +40.25	43.71 +14.84
	Qwen2.5-VL-7B	74.15	74.14	58.90	44.88
	Qwen2.5-VL-7B + GRPO	98.73 +24.58	98.73 +24.59	86.02 +27.12	66.27 +21.39

vs. 77.54%), but its 8-class Macro-F1 improvement under GRPO is far smaller (3B: +3.42 vs. +25.75 pp). This reflects a fundamental information-theoretic asymmetry between modalities—text carries rich semantic cues (specific intimidation phrases, inducement patterns) that support fraud-type discrimination, whereas visual features, though sufficient for anomaly detection, are largely category-agnostic across fraud types.

The image–text modality yields the largest improvements on both models, with the 7B model reaching Macro-F1 = 66.27% (+21.39 pp)—the best configuration overall. Notably, the 7B baseline 2-class accuracy on image–text (74.15%) is *lower* than both its text-only (78.81%) and image-only (86.02%) baselines, yet after GRPO training it

reaches 98.73%, the highest across all settings. This paradox reveals that the untrained MLLM suffers from attention dispersion when processing multimodal inputs and fails to integrate cross-modal information effectively. GRPO training resolves the issue by learning a reward-guided co-reasoning strategy that harnesses the complementary strengths of both modalities.

5.4. Ablation: scene-reward weight λ

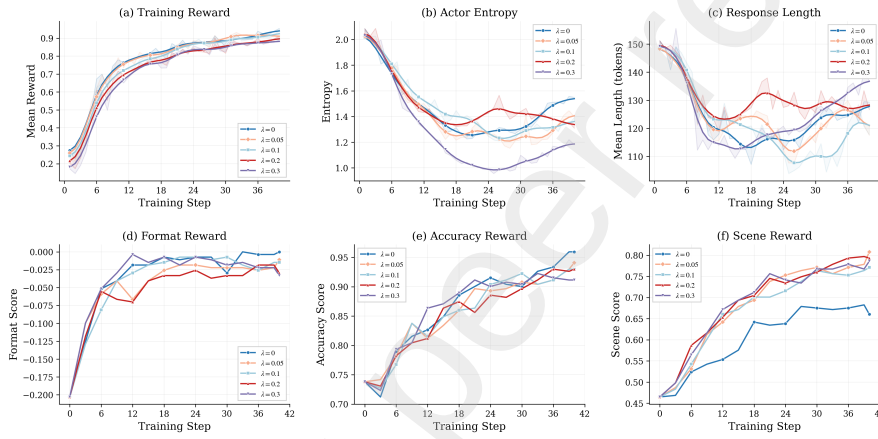


Fig. 9: Training dynamics under different scene-reward weights λ . (a) Total reward; (b) policy entropy; (c) response length; (d) format reward; (e) accuracy reward; (f) scene reward.

Table 4: Ablation on the scene-reward weight λ (Qwen2.5-VL-3B).

λ	2-Class		8-Class	
	Acc	F1	Acc	Macro-F1
0	95.76	95.76	65.54	40.26
0.05	96.19	96.19	81.50	52.54
0.1	96.47	96.54	78.25	45.13
0.2	95.06	95.05	78.95	44.79
0.3	95.90	95.97	78.53	46.75

To investigate the effect of λ we train Qwen2.5-VL-3B with $\lambda \in \{0, 0.05, 0.1, 0.2, 0.3\}$. Fig. 9 shows the training dynamics. All configurations converge to total rewards above 0.9 (Fig. 9a), and the format-reward differences are negligible (Fig. 9d), confirming that λ does not affect basic convergence or format learning.

The critical distinction lies in the accuracy–scene reward trade-off. The accuracy reward (Fig. 9e) decreases mildly as λ grows (0.959 at $\lambda = 0$ vs. 0.930 at $\lambda = 0.2$). In contrast, the scene reward (Fig. 9f) shows a dramatic *non-linear* response: at $\lambda = 0$ the scene reward plateaus at 0.66, but even a minimal increase to $\lambda = 0.05$ triggers a jump to 0.81. Without any scene supervision, GRPO’s advantage estimation provides no effective gradient for scene classification—the model’s scene-recognition ability then arises solely from indirect spillover of binary-classification training. Introducing even a small scene-reward weight injects a discriminative signal into the advantage estimation, fundamentally altering the model’s learning behaviour.

Table 4 confirms these observations on the test set. Binary metrics fluctuate by less than 1.5 pp across all settings, confirming minimal interference with the primary task. The 8-class metrics exhibit a clear inverted-U pattern: accuracy jumps from 65.54% ($\lambda = 0$) to 81.50% ($\lambda = 0.05$), then retreats to 78–79% for $\lambda \geq 0.1$. This non-monotonic behaviour reflects a multi-task learning tension: when λ is too large, the model prioritises scene correctness and may sacrifice binary-judgement certainty.

Overall, $\lambda = 0.05$ achieves the best or near-best performance on every metric, making it the recommended configuration. This ablation shows that the core value of the scene reward lies in injecting a scene-discriminative signal into the advantage estimation, rather than in amplifying its weight: moderate auxiliary supervision efficiently activates fine-grained recognition.

5.5. Qualitative analysis

To provide intuitive insight into how GRPO training improves multimodal fraud detection, we present two representative case studies—one legitimate message and

one fraudulent message—comparing the chain-of-thought (CoT) reasoning and final predictions of the baseline Qwen2.5-VL-7B model with those of its GRPO-trained counterpart.

Table 5: Case 1 — legitimate message (loyalty-points campaign). The baseline misclassifies it as fraud by relying on shallow text heuristics, whereas the GRPO-trained model correctly integrates visual evidence.

✗ incorrect; ✓ correct. Key evidence in the CoT reasoning is highlighted in teal.

Input Image	Model	CoT Reasoning (abbrev.)	Auth.	Scene
	Qwen2.5-VL-7B (Baseline)	“The message begins with ‘Dear customer’ but provides no specific company name... the codes appear randomly generated... the message may be suspected of fraud.”	Fraud ✗	Fake task rewards ✗
	Qwen2.5-VL-7B + GRPO	“ The image shows a loyalty-points campaign... usage venues include carrier stores, KFC and a cinema chain... no requests for money or personal information... a normal promotional campaign.”	Legit. ✓	Legit. ✓

Analysis of Case 1. The input (Table 5) is a legitimate promotional MMS from a mobile carrier advertising a loyalty-points redemption campaign; the text describes how users may send specific codes to redeem electronic vouchers. The baseline model fixates on superficial textual cues—the absence of a specific company name and the observation that the redemption codes “appear randomly generated”—and incorrectly classifies the message as fraudulent. This reveals a core limitation: the baseline relies on shallow text heuristics without leveraging the image, which clearly depicts a legitimate carrier promotion with specific partner logos and redemption venues.

In contrast, the GRPO-trained model integrates both modalities effectively. It identifies the image as a loyalty-points campaign, notes the verifiable usage venues (carrier stores, phone retailers, KFC and a cinema chain) and observes the absence of requests for monetary transactions or personal information. This cross-modal co-reasoning leads to the correct prediction. The case demonstrates how GRPO training enables the model to overcome text-level ambiguity by incorporating corroborating visual evidence.

Table 6: Case 2 — fake-loans fraud message. The baseline fails to recognise phishing indicators, whereas the GRPO-trained model identifies fraud-specific operational tactics. ✗ incorrect; ✓ correct. Key evidence in the CoT reasoning is highlighted in teal.

Input Image	Model	CoT Reasoning (abbrev.)	Auth.	Scene
	Qwen2.5-VL-7B (Baseline)	“The message mentions an anomaly risk in a financial account... it provides two QR codes and a phone number... this looks like a normal account security notification.”	Legit. ✗	Legit. ✗
	Qwen2.5-VL-7B + GRPO	“The message claims a problem with the recipient’s credit account... it instructs downloading screen-mirroring apps, common tools used by scammers for remote control... QR codes and a phone number are typical phishing tactics.”	Fraud ✓	Fake loans ✓

Analysis of Case 2. The input (Table 6) is a fraudulent MMS impersonating a major e-commerce platform’s financial service. The image contains fabricated security notices with QR codes and a phone number, while the text instructs the recipient to install third-party applications for “identity verification and account recovery”. The baseline interprets the message as “a normal account security notification”, missing the telltale signs of a phishing attack—the request to install unrelated third-party meeting and screen-mirroring software, the use of QR codes to redirect victims to controlled channels, and the urgency-inducing framing.

The GRPO-trained model correctly identifies the message as fraudulent and assigns it to the “fake loans” category. Its reasoning chain explicitly flags the instruction to download screen-mirroring apps as “common tools used by scammers to gain remote control” and identifies the QR codes as “typical phishing tactics”. This demonstrates that GRPO training, guided by the composite reward, equips the model with domain-specific knowledge about fraud patterns that goes beyond surface-level text comprehension and encompasses an understanding of the operational tactics employed by different scammers.

Summary of qualitative findings. These two cases collectively illustrate the core challenges addressed by our approach. First, multimodal fraud detection re-

quires genuine cross-modal reasoning: the baseline tends to process text and image independently, producing errors when either modality alone is ambiguous. Second, fine-grained fraud-type classification demands understanding of scam-specific operational patterns (e.g., screen-sharing tools, QR-code redirection) that cannot be learned through standard instruction tuning alone. The scene-reward component in our composite reward directly incentivises the model to acquire such discriminative knowledge during GRPO training.

6. Conclusion

We have presented GRPO-MMS, a reinforcement-learning framework for multimodal fraud message detection. Our contributions are threefold. (i) We release MMS-VQA, a multimodal benchmark spanning eight categories and three input modalities with a generator-based evaluation protocol. (ii) We design a three-component composite reward integrating format, authenticity and scene signals for GRPO-based training. (iii) We present comprehensive experiments demonstrating substantial improvements (up to +31.36 pp in 8-class accuracy), with the image-text modality yielding the largest gains.

Our ablations reveal that the primary role of the scene reward in GRPO is to inject scene-discriminative gradients into the advantage estimation; even a minimal weight ($\lambda = 0.05$) is sufficient to activate fine-grained classification. The qualitative analysis further shows that GRPO training enables genuine cross-modal reasoning—the model learns to integrate visual evidence with textual analysis rather than to rely on shallow single-modality heuristics. These findings have broader implications for reward-function design in RL-based pattern recognition: carefully crafted auxiliary rewards can efficiently unlock multi-granularity reasoning capabilities in multimodal models.

Future work will explore larger-scale datasets with more diverse fraud categories, incorporate the temporal dynamics of evolving fraud patterns, and investigate the transferability of the learned models to real-world deployment scenarios.

CRedit authorship contribution statement

Zhengchao Huang: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Validation, Visualization, Writing — original draft. **Jiaran Li:** Software, Data curation, Validation, Writing — review & editing. **Wenming Yang:** Conceptualization, Supervision, Resources, Project administration, Funding acquisition, Writing — review & editing.

Data availability

The MMS-VQA dataset and the code of GRPO-MMS will be released to the research community upon acceptance of this manuscript. All underlying SMS texts are derived from a publicly available telecom-fraud SMS corpus [11]; synthetic images are generated from the same public sources using openly available text-to-image models.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used generative-AI tools solely for language polishing and grammar checking. After using these tools, the authors carefully reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

- [1] L. Jiang, C. Zhang, X. Qin, Y. Zhou, G. Huang, H. Li, J. Li, Telecom fraud recognition based on large language model neuron selection, Mathematics 13 (11) (2025) 1826. [doi:10.3390/math13111826](https://doi.org/10.3390/math13111826).
- [2] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, BERT: Pre-training of deep bidirectional transformers for language understanding, in: Proceedings of the 2019

- Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), 2019, pp. 4171–4186. [doi:10.18653/v1/N19-1423](https://doi.org/10.18653/v1/N19-1423).
- [3] P. Hájek, R. Henriques, Mining corporate annual reports for intelligent detection of financial statement fraud – a comparative study of machine learning methods, *Knowledge-Based Systems* 128 (2017) 139–152. [doi:10.1016/j.knosys.2017.05.001](https://doi.org/10.1016/j.knosys.2017.05.001).
- [4] S. Mishra, D. Soni, SMS phishing dataset for machine learning and pattern recognition, in: *Soft Computing and its Engineering Applications (icSoftComp 2022)*, Vol. 1788 of *Communications in Computer and Information Science*, Springer, 2023, pp. 684–694. [doi:10.1007/978-3-031-27524-1_57](https://doi.org/10.1007/978-3-031-27524-1_57).
- [5] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, B. Ommer, High-resolution image synthesis with latent diffusion models, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 10684–10695. [doi:10.1109/CVPR52688.2022.01042](https://doi.org/10.1109/CVPR52688.2022.01042).
- [6] A. Ramesh, P. Dhariwal, A. Nichol, C. Chu, M. Chen, Hierarchical text-conditional image generation with CLIP latents, *arXiv preprint arXiv:2204.06125* (2022).
- [7] Black Forest Labs, FLUX: Frontier text-to-image generation models, <https://bfl.ai/models/flux-2>, accessed: 2026-04-18 (2025).
- [8] ByteDance Seed, Seedream 4.0: Toward next-generation multimodal image generation, *arXiv preprint arXiv:2509.20427* (2025).
- [9] X. Zhou, J. Wu, R. Zafarani, SAFE: Similarity-aware multi-modal fake news detection, in: *Advances in Knowledge Discovery and Data Mining (PAKDD 2020)*, Vol. 12085 of *Lecture Notes in Computer Science*, Springer, 2020, pp. 354–367. [doi:10.1007/978-3-030-47436-2_27](https://doi.org/10.1007/978-3-030-47436-2_27).

- [10] Y. Wu, P. Zhan, Y. Zhang, L. Wang, Z. Xu, Multimodal fusion with co-attention networks for fake news detection, in: Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021, 2021, pp. 2560–2569. [doi:10.18653/v1/2021.findings-acl.226](https://doi.org/10.18653/v1/2021.findings-acl.226).
- [11] Y. Zhou, Y. Yang, Q. Ying, Z. Qian, X. Zhang, Multimodal fake news detection via CLIP-guided learning, in: Proceedings of the IEEE International Conference on Multimedia and Expo (ICME), 2023, pp. 2825–2830. [doi:10.1109/ICME55011.2023.00480](https://doi.org/10.1109/ICME55011.2023.00480).
- [12] Y. Chen, D. Li, P. Zhang, J. Sui, Q. Lv, L. Tun, L. Shang, Cross-modal ambiguity learning for multimodal fake news detection, in: Proceedings of the ACM Web Conference (WWW), 2022, pp. 2897–2905. [doi:10.1145/3485447.3511968](https://doi.org/10.1145/3485447.3511968).
- [13] M. S. Rana, M. N. Nobi, B. Murali, A. H. Sung, Deepfake detection: A systematic literature review, IEEE Access 10 (2022) 25494–25513. [doi:10.1109/ACCESS.2022.3154404](https://doi.org/10.1109/ACCESS.2022.3154404).
- [14] Z. Wang, Y. Guo, W. Zuo, Deepfake forensics via an adversarial game, IEEE Transactions on Image Processing 31 (2022) 3541–3552. [doi:10.1109/TIP.2022.3172845](https://doi.org/10.1109/TIP.2022.3172845).
- [15] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat, et al., GPT-4 technical report, arXiv preprint arXiv:2303.08774 (2023).
- [16] H. Liu, C. Li, Q. Wu, Y. J. Lee, Visual instruction tuning, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 36, 2023, pp. 34892–34916.
- [17] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, K. Dang, P. Wang, S. Wang, J. Tang, et al., Qwen2.5-VL technical report, arXiv preprint arXiv:2502.13923 (2025).

- [18] S. Antol, A. Agrawal, J. Lu, M. Mitchell, D. Batra, C. L. Zitnick, D. Parikh, VQA: Visual question answering, in: Proceedings of the IEEE International Conference on Computer Vision (ICCV), 2015, pp. 2425–2433. [doi:10.1109/ICCV.2015.279](https://doi.org/10.1109/ICCV.2015.279).
- [19] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, D. Zhou, Chain-of-thought prompting elicits reasoning in large language models, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 35, 2022, pp. 24824–24837.
- [20] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. L. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, J. Schulman, J. Hilton, F. Kelton, L. Miller, M. Simens, A. Askell, P. Welinder, P. F. Christiano, J. Leike, R. Lowe, Training language models to follow instructions with human feedback, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 35, 2022, pp. 27730–27744.
- [21] Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, X. Bi, H. Zhang, M. Zhang, Y. K. Li, Y. Wu, et al., DeepSeekMath: Pushing the limits of mathematical reasoning in open language models, arXiv preprint arXiv:2402.03300 (2024).
- [22] Z. Wei, H. Pan, L. Qiao, X. Niu, P. Dong, D. Li, Cross-modal knowledge distillation in multi-modal fake news detection, in: Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2022, pp. 4733–4737. [doi:10.1109/ICASSP43922.2022.9747280](https://doi.org/10.1109/ICASSP43922.2022.9747280).
- [23] J. Jing, H. Wu, J. Sun, X. Fang, H. Zhang, Multimodal fake news detection via progressive fusion networks, Information Processing & Management 60 (1) (2023) 103120. [doi:10.1016/j.ipm.2022.103120](https://doi.org/10.1016/j.ipm.2022.103120).
- [24] C. Miao, Z. Tan, Q. Chu, N. Yu, G. Guo, Hierarchical frequency-assisted interactive networks for face manipulation detection, IEEE Transactions on Informa-

- tion Forensics and Security 17 (2022) 3008–3021. [doi:10.1109/TIFS.2022.3198275](https://doi.org/10.1109/TIFS.2022.3198275).
- [25] L. Shi, J. Zhang, Z. Ji, J. Bai, S. Shan, Real face foundation representation learning for generalized deepfake detection, Pattern Recognition 161 (2025) 111299. [doi:10.1016/j.patcog.2024.111299](https://doi.org/10.1016/j.patcog.2024.111299).
- [26] J. Li, D. Li, S. Savarese, S. C. H. Hoi, BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models, in: Proceedings of the 40th International Conference on Machine Learning (ICML), 2023, pp. 19730–19742.
- [27] Y. Li, Y. Zhang, C. Wang, Z. Zhong, Y. Chen, R. Chu, S. Liu, J. Jia, Mini-Gemini: Mining the potential of multi-modality vision language models, arXiv preprint arXiv:2403.18814 (2024).
- [28] Z. Hu, P. Yang, Y. Jiang, Z. Bai, Prompting large language model with context and pre-answer for knowledge-based VQA, Pattern Recognition 151 (2024) 110399. [doi:10.1016/j.patcog.2024.110399](https://doi.org/10.1016/j.patcog.2024.110399).
- [29] H. A. Alawwad, A. Alhothali, U. Naseem, A. Alkhathlan, A. Jamal, Enhancing textual textbook question answering with large language models and retrieval augmented generation, Pattern Recognition 162 (2025) 111332. [doi:10.1016/j.patcog.2024.111332](https://doi.org/10.1016/j.patcog.2024.111332).
- [30] T. Baltrušaitis, C. Ahuja, L.-P. Morency, Multimodal machine learning: A survey and taxonomy, IEEE Transactions on Pattern Analysis and Machine Intelligence 41 (2) (2019) 423–443. [doi:10.1109/TPAMI.2018.2798607](https://doi.org/10.1109/TPAMI.2018.2798607).
- [31] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, I. Sutskever, Learning transferable visual models from natural language supervision, in: Proceedings of the 38th International Conference on Machine Learning (ICML), 2021, pp. 8748–8763.

- [32] P. Xu, X. Zhu, D. A. Clifton, Multimodal learning with transformers: A survey, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45 (10) (2023) 12113–12132. [doi:10.1109/TPAMI.2023.3275156](https://doi.org/10.1109/TPAMI.2023.3275156).
- [33] J. Zhang, K. He, Z. Feng, S. Sun, X. Sun, Z. Yuan, J. Yu, Self-adaptive image-text fusion for medical image classification, *Pattern Recognition* 167 (2025) 111715. [doi:10.1016/j.patcog.2025.111715](https://doi.org/10.1016/j.patcog.2025.111715).
- [34] L. Chen, J. Li, X. Dong, P. Zhang, Y. Zang, Z. Chen, H. Duan, J. Wang, Y. Qiao, D. Lin, F. Zhao, Are we on the right way for evaluating large vision-language models?, *Advances in Neural Information Processing Systems (NeurIPS)* 37 (2024).
- [35] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, Proximal policy optimization algorithms, *arXiv preprint arXiv:1707.06347* (2017).
- [36] D. Guo, D. Yang, H. Zhang, J. Song, R. Zhang, R. Xu, Q. Zhu, S. Ma, P. Wang, X. Bi, et al., DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning, *arXiv preprint arXiv:2501.12948* (2025).
- [37] G. Sheng, C. Zhang, Z. Ye, X. Wu, W. Zhang, R. Zhang, Y. Peng, H. Lin, C. Wu, HybridFlow: A flexible and efficient RLHF framework, in: *Proceedings of the Twentieth European Conference on Computer Systems (EuroSys)*, 2025, pp. 1279–1297. [doi:10.1145/3689031.3696075](https://doi.org/10.1145/3689031.3696075).